MILITARY SPECIFICATION SHEET

TRANSFORMERS, POWER, STEPDOWN AND STEP-UP,
Ⓐ
TF5S03ZZ AND TF5T03ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformers described herein shall consist of this document and the latest issue of Specification MIL-T-27.

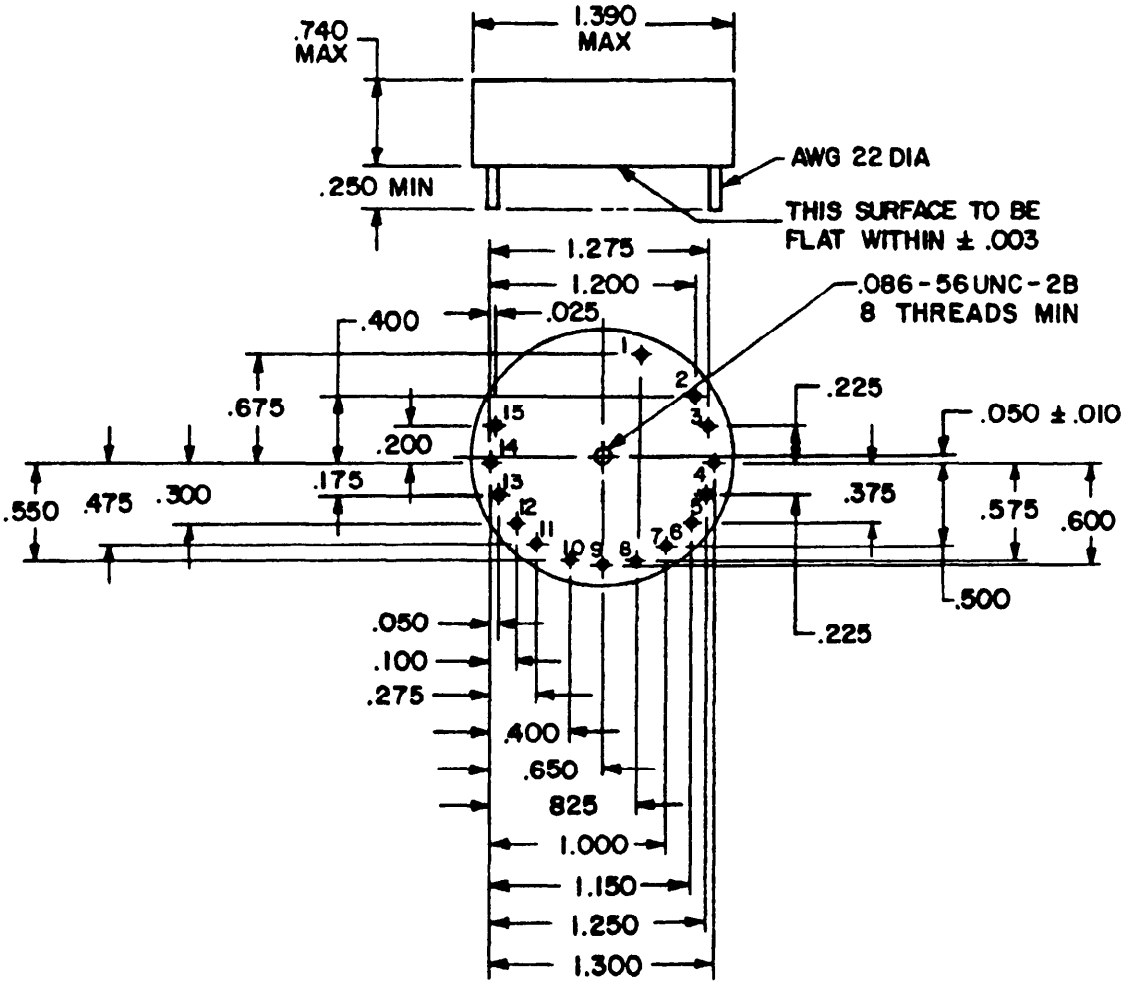


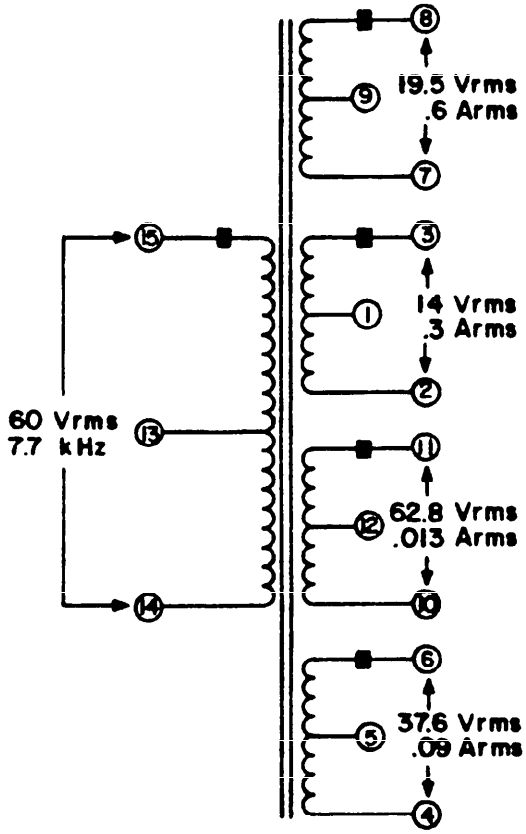

FIGURE 1. Dimensions and configuration.

Ⓐ denotes change



FSC 5950

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WORKING VOLTAGE: 175 V MAX

CIRCUIT DIAGRAM AND MARKING

| INCHES | MM | INCHES | MM | INCHES | MM |
|---|---|---|---|---|---|
| .003 | .08 | .275 | 6.98 | .675 | 17.14 |
| .010 | .25 | .300 | 7.62 | .740 | 18.80 |
| .025 | .64 | .375 | 9.52 | .825 | 20.96 |
| .050 | 1.27 | .400 | 10.16 | 1.000 | 25.40 |
| .086 | 2.18 | .475 | 12.06 | 1.150 | 29.21 |
| .100 | 2.54 | .500 | 12.70 | 1.200 | 30.48 |
| .175 | 4.44 | .550 | 13.97 | 1.250 | 31.75 |
| .200 | 5.08 | .575 | 14.60 | 1.275 | 32.38 |
| .225 | 5.72 | .600 | 15.24 | 1.300 | 33.02 |
| .250 | 6.35 | .650 | 16.51 | 1.390 | 35.31 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm).
4. Marking shall be on the top of the case.

FIGURE 1. Dimensions and configuration - Continued.



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REQUIREMENTS: (When numbers in parentheses, i.e. (1-2) are used, they indicate the windings and the extreme terminals of the windings.)

Electrical ratings:
Primary voltage (14-15): 60 volts rms, 7.7 kilohertz.
Primary current: .34 ampere rms.
Primary power: 20.4 volts, ampere.
Secondary voltage:
(7-8): 19.5 volts rms, .6 ampere rms.
(2-3): 14 volts rms, .3 ampere rms.
(10-11): 62.8 volts rms, .013 ampere rms.
(4-6): 37.6 volts rms, .09 ampere rms.
Working voltage: 175 volts, maximum.
DC resistance:
(14-15): .4 ohm.
(7-8): .24 ohm.
(2-3); .35 ohm.
(4-6): 1.16 ohms.
(10-11): 10 ohms.

Design and construction:
Dimensions and configuration: See figure 1.
Duty cycle: Continuous.
Case: Encapsulated.
Material: Epoxy.
Terminals: Pin type.
Material: Type S, tin lead or lead-tin coating in accordance with QQ-W-343.
Height: .250 inch, minimum.
Diameter: AWG 22 (.025 inch)
Weight: 4 ounces, maximum.
(A) Operating temperature range: see table I.

Terminal strength: MIL-STD-202, method 211, test condition A, (A) 2 pounds.

Dielectric withstanding voltage (at sea level): 1,000 volts rms.

Electrical characteristics:
No load: With 60 volts rms and 7.7 kilohertz in (14-15):
Current in (14-15): 44 milliamperes, maximum.
Power in (14-15): 2.3 watts, maximum.
Voltage across (7-8): 19.5 volts rms ±1 percent.
Voltage across (2-3): 14 volts rms ±1 percent.
Voltage across (10-11): 62.8 volts rms ±1 percent.
Voltage across (4-6): 37.6 volts rms ±1 percent.
Rated load: With 60 volts rms, and 7.7 kilohertz (14-15):
Voltage across (7-8): 19.5 volts rms ±5 percent.
Voltage across (2-3): 14 volts rms ±5 percent.
Voltage across (10-11): 62.8 volts rms ±5 percent.
Voltage across (4-6): 37.6 volts rms ±5 percent.
Polarity: Additive with terminals 3 and 7, 11 and 2, and 6 and 10 connected.

(A) Temperature rise: 40°C with 60 volts rms, 7.7 kilohertz across (14-15) at specified ambient temperature (see table I).

Vibration (high frequency): MIL-STD-202, method 204.

Shock (specified pulse): MIL-STD-202, method 213, test condition A.

Marking location: See figure 1.

Part number: M27/194-(dash number from table I).



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(A) TABLE I. Temperature characteristics.

| Dash no. | Type designation | Ambient temperature | Temperature rise | Operating temperature range |
|---|---|---|---|---|
| 01 | TF5S03ZZ | 90°C | 40°C | -55° to +130°C |
| 02 | TF5T03ZZ | 130°C | 40°C | -55° to +170°C |

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME, AR,
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - EL

Agent:
DLA - ES

(Project 5950-0556-8)



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DOCUMENT IDENTIFIER AND TITLE MIL-T-27/194A TRANSFORMERS, POWER, STEPDOWN AND STEP-UP, TF5S03ZZ AND TF5T03ZZ

| NAME OF ORGANIZATION AND ADDRESS | CONTRACT NUMBER |
|---|---|
| | MATERIAL PROCURED UNDER A<br>☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT |

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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